SUPPORTING ADULT GUIDANCE

Solving linear inequalities

Inequalities · Lesson 2. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson teaches learners to **solve linear inequalities** by inverse operations, and to **discover** why multiplying or dividing by a negative number reverses the inequality. Learners predict and then see the reversal on a number line before they use it, solve on a balance drawn for a test value, and check working by testing values line by line.

Driving Question: *Can I solve an inequality like an equation — and where does it behave differently?*

The mathematics, in plain terms

Adding or subtracting the same number, or multiplying or dividing by the same positive number, preserves an inequality. Multiplying or dividing both sides by a negative number reverses it, because the map $t \to -t$ reflects the number line about 0 and reverses order: $a < b$ implies $-a > -b$. A solution can be checked by substituting a value from inside the solution set (the original statement is true) and one from outside (it is false).

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Two steps, no reversal

Solve $3x + 2 > 11$.

1. Subtract 2 from both sides.	$3x > 9$
2. Divide both sides by 3. It is positive, so the symbol stays.	$x > 3$
3. Check with $x = 4$: $3 \times 4 + 2 = 14$, and $14 > 11$.	true ✓

$x > 3$. Open circle at 3, arrow to the right.

Example 2 — When the symbol reverses

Solve $5 - 2x \geq 1$.

1. Subtract 5 from both sides.	$-2x \geq -4$
2. Divide both sides by –2. It is negative, so reverse ≥ to ≤.	$x \leq 2$
3. Check with $x = 0$: $5 - 0 = 5$, and $5 \geq 1$.	true ✓

$x \leq 2$. **Filled circle at 2, arrow to the left.**

Discover before you use it

Resist stating the reversal rule at the start. In Connection, learners predict whether an operation keeps the order of two numbers, then watch. Most will discover the rule themselves. In the balance activity, the balance is drawn for one value that satisfies the inequality, so a reversal is visible as the balance tipping the other way. When a learner forgets to reverse, ask them to test a value rather than correcting them.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Predict whether an operation keeps the order of two numbers, then watch it on a number line.	~8 min
Movement	The tipped balance: solve inequalities step by step, choosing the symbol after a negative step.	~14 min
Reflection	Test someone's working line by line and find the line that goes wrong.	~10 min
Creativity	Form and solve inequalities from real situations, and interpret the answer.	~8 min
Rest	Picture the number line flipping over zero.	~2 min
Nutrition	Intellectual: asking why a rule is true, and checking for yourself.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Reversing the symbol whenever there is a minus sign anywhere.
The reversal belongs only to multiplying or dividing by a negative. Ask: 'Did you multiply or divide by a negative number in this step?' Subtracting 5 does not reverse.

Forgetting to reverse when dividing by a negative.
Ask for a test value: 'Is $x = 0$ in your answer? Does it make the original inequality true?' The disagreement shows the slip.

Treating the answer as one number ($x = 3$).
'Is 4 a solution? Is 10? So the answer is not one value — how do we write all of them?'

Losing the meaning in context (8.5 taxi miles).
Return to the story: 'Can the meter charge for half a mile? What is the greatest whole number that keeps within the limit?'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: solving one- and two-step equations, and the inequality symbols from Lesson 1. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner writes $-2x \geq -4$, so $x \geq 2$. The aim is a question that lets them find the slip themselves.

- "Pick a value in your answer, say $x = 3$. Is $5 - 2 \times 3 \geq 1$ true?"

- “What did you divide by in the last step? Was it positive or negative?”
- “What happens to $2 < 5$ if you multiply both sides by -1?”
- “Which value of x would make both sides equal? Is that value in or out?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why not just teach the rule?

Because a rule learned as a trick is easily misapplied (for example, reversing after subtracting). Seeing the reflection makes it clear exactly when the reversal applies.

Can learners avoid dividing by a negative?

Yes: they can add the x term to the other side first ($5 - 2x \geq 1$ becomes $5 \geq 1 + 2x$). Both methods are correct; seeing both deepens understanding.

Do answers have to be whole numbers?

No. $3 + 2m \leq 20$ gives $m \leq 8.5$. Only the context decides whether whole numbers are needed.

What comes next?

Lesson 3 represents solution sets on a number line in more detail, including double-ended inequalities, and reads inequalities back from a number line.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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